

NETWORK SERVICES LAB

Windows Server 2022: DHCP · DNS · IIS · Virtual Printer

Lab Activity Sheet — Student Group Edition

Duration	2 hours (recommended pace shown in schedule)
Platform	Windows Server 2022 (VM on group laptop)
Services	DHCP Server, DNS Server, IIS Web Server, Print Services
Network	2 VLANs per group — inter-VLAN routing via router/gateway
Objective	Configure all four services so clients on both VLANs receive IPs, resolve names, browse a web page, and print.

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Section 1 — Network Address Reference

The table below lists the complete IP plan for all 8 VLANs across 4 groups. Locate your group's two VLANs and keep this table visible throughout the lab.

VLAN ID	VLAN Name	Network/CIDR	Subnet Mask	Gateway	Server IP	DHCP Range	Hosts
VLAN 10	Guest	10.10.0.0/26	255.255.255.192	10.10.0.1	10.10.0.62	10.10.0.2 – 10.10.0.61	62
VLAN 20	IT	10.10.0.64/27	255.255.255.224	10.10.0.65	10.10.0.94	10.10.0.66 – 10.10.0.93	30
VLAN 30	SoICT	10.10.0.96/27	255.255.255.224	10.10.0.97	10.10.0.126	10.10.0.98 – 10.10.0.125	30
VLAN 40	HR	10.10.0.128/27	255.255.255.224	10.10.0.129	10.10.0.158	10.10.0.130 – 10.10.0.157	30
VLAN 50	Engineering	10.10.0.160/27	255.255.255.224	10.10.0.161	10.10.0.190	10.10.0.162 – 10.10.0.189	30
VLAN 60	Finance	10.10.0.192/28	255.255.255.240	10.10.0.193	10.10.0.206	10.10.0.194 – 10.10.0.205	14
VLAN 70	Printers & IoT	10.10.0.208/28	255.255.255.240	10.10.0.209	10.10.0.222	10.10.0.210 – 10.10.0.221	14
VLAN 80	Management	10.10.0.224/28	255.255.255.240	10.10.0.225	10.10.0.238	10.10.0.226 – 10.10.0.237	14

KEY RULE

First usable IP of each VLAN = Default Gateway (configured on the router sub-interface).
 Last usable IP of each VLAN = Windows Server 2022 static IP (your group's server).
 All other usable IPs between first+1 and last-1 form the DHCP pool.

Section 2 — Group IP Assignments

Each group manages two VLANs. Find your group below. Use only the addresses shown for your group.

Group 1 | VLAN 10 (Guest) + VLAN 20 (IT)

VLAN	Network/CIDR	Subnet Mask	Gateway (your router)	Server IP (static)	DHCP Pool	Hosts
VLAN 10 – Guest	10.10.0.0/26	255.255.255.192	10.10.0.1	10.10.0.62	10.10.0.2 – 10.10.0.61	62
VLAN 20 – IT	10.10.0.64/27	255.255.255.224	10.10.0.65	10.10.0.94	10.10.0.66 – 10.10.0.93	30

Group 2 | VLAN 30 (SoICT) + VLAN 40 (HR)

VLAN	Network/CIDR	Subnet Mask	Gateway (your router)	Server IP (static)	DHCP Pool	Hosts
VLAN 30 – SoICT	10.10.0.96/27	255.255.255.224	10.10.0.97	10.10.0.126	10.10.0.98 – 10.10.0.125	30
VLAN 40 – HR	10.10.0.128/27	255.255.255.224	10.10.0.129	10.10.0.158	10.10.0.130 – 10.10.0.157	30

Group 3 | VLAN 50 (Engineering) + VLAN 60 (Finance)

VLAN	Network/CIDR	Subnet Mask	Gateway (your router)	Server IP (static)	DHCP Pool	Hosts
VLAN 50 – Engineering	10.10.0.160/27	255.255.255.224	10.10.0.161	10.10.0.190	10.10.0.162 – 10.10.0.189	30
VLAN 60 – Finance	10.10.0.192/28	255.255.255.240	10.10.0.193	10.10.0.206	10.10.0.194 – 10.10.0.205	14

Group 4 | VLAN 70 (Printers & IoT) + VLAN 80 (Management)

VLAN	Network/CIDR	Subnet Mask	Gateway (your router)	Server IP (static)	DHCP Pool	Hosts
VLAN 70 – Printers & IoT	10.10.0.208/28	255.255.255.240	10.10.0.209	10.10.0.222	10.10.0.210 – 10.10.0.221	14
VLAN 80 – Management	10.10.0.224/28	255.255.255.240	10.10.0.225	10.10.0.238	10.10.0.226 – 10.10.0.237	14

Section 3 — Recommended Time Schedule

Follow this pacing to complete all tasks within 2 hours. The instructor may adjust times.

Time	Task	Deliverable
0:00 – 0:10	Read lab sheet, study IP plan, verify physical connections	Understand topology
0:10 – 0:25	Configure Windows Server 2022 static IP (both VLANs via VM dual NIC or VLAN tagging)	Server responds to ping from gateway
0:25 – 0:50	Install & configure DHCP Server role (scope for each VLAN)	Clients receive IPs automatically
0:50 – 1:10	Install & configure DNS Server role (forward lookup zone + A records)	nslookup resolves lab hostnames
1:10 – 1:30	Install IIS Web Server – publish a custom default page	Clients browse to server IP in browser
1:30 – 1:50	Install Print Services role – add virtual printer	Clients can connect and print a test page
1:50 – 2:00	Final testing, peer verification & cleanup	Instructor sign-off sheet completed

TIP

Work in parallel where possible: one student configures the server role while another tests from the client laptop.

Section 4 — Lab Tasks (Step-by-Step)

Complete each task in order. Use your group's IP information from Section 2 wherever you see [YOUR VALUES].

Task A — Assign a Static IP to Windows Server 2022

Your server must have a fixed IP address (the last usable address of each VLAN). Since you have two VLANs, configure two network adapters in your VM — one per VLAN — or use VLAN tagging if your hypervisor supports it.

Step A.1: Open Network Adapter Settings

1. Log in to Windows Server 2022.
2. Right-click the Start button → select Network Connections.
3. Click Change adapter options.
4. You should see two Ethernet adapters (one per VLAN). Rename them VLAN-XX and VLAN-YY using your VLAN IDs for clarity.

NOTE

If only one adapter is visible, add a second virtual NIC in your hypervisor settings (VMware: VM Settings → Add → Network Adapter; VirtualBox: Settings → Network → Adapter 2).

Step A.2: Configure VLAN 1 Adapter (first VLAN of your group)

5. Right-click the first adapter → Properties.
6. Select Internet Protocol Version 4 (TCP/IPv4) → Properties.
7. Choose Use the following IP address.
8. IP address: [last usable IP of your first VLAN]
9. Subnet mask: [mask of your first VLAN]
10. Default gateway: [first usable IP = gateway of your first VLAN]
11. Preferred DNS server: 127.0.0.1 (server will be its own DNS)
12. Click OK → Close.

IMPORTANT

Do not use DHCP for the server itself. It must always have the same static IP.

Step A.3: Configure VLAN 2 Adapter (second VLAN of your group)

13. Right-click the second adapter → Properties → IPv4 → Properties.
14. IP address: [last usable IP of your second VLAN]
15. Subnet mask: [mask of your second VLAN]
16. Default gateway: [first usable IP = gateway of your second VLAN]
17. Preferred DNS server: 127.0.0.1
18. Click OK → Close.

Step A.4: Verify Connectivity

19. Open Command Prompt (cmd.exe) as Administrator.
20. Ping the gateway of your first VLAN: ping [gateway IP 1]
21. Ping the gateway of your second VLAN: ping [gateway IP 2]
22. Both should reply. If not, check the physical cable and VLAN trunk configuration on your switch port.

NOTE

If pings fail, verify the VM's virtual switch is mapped to the correct physical port or VLAN.

Task B — Install and Configure DHCP Server

Step B.1: Install the DHCP Server Role

23. Open Server Manager → click Manage → Add Roles and Features.
24. Click Next through the wizard until Roles.
25. Check DHCP Server → click Add Features when prompted.
26. Click Next → Next → Install.
27. Wait for installation to complete, then click Close.
28. In Server Manager, click the yellow flag notification → Complete DHCP Configuration → Commit → Close.

Step B.2: Create DHCP Scope for VLAN 1

29. Open DHCP (Server Manager → Tools → DHCP).
30. Expand your server name → right-click IPv4 → New Scope.
31. Name: VLAN-[ID]-[Name] (example: VLAN-10-Guest).
32. Start IP: [first+1 IP of VLAN 1] End IP: [last-1 IP of VLAN 1].
33. Subnet mask: [mask of VLAN 1] → Next.
34. Exclusions: none required (pool already excludes gateway and server).
35. Lease duration: 8 hours → Next.
36. Configure options NOW → Next.
37. Router (option 003): enter [gateway IP of VLAN 1] → Add → Next.
38. DNS Server (option 006): enter [server IP of VLAN 1] → Add → Next.
39. Activate the scope: Yes → Finish.

NOTE

The DHCP pool must NOT include the gateway (first usable) or the server (last usable). Your start IP = gateway+1, end IP = server-1.

Step B.3: Create DHCP Scope for VLAN 2

40. Right-click IPv4 → New Scope again.
41. Name: VLAN-[ID]-[Name] (example: VLAN-20-IT).
42. Start IP: [first+1 IP of VLAN 2] End IP: [last-1 IP of VLAN 2].
43. Subnet mask: [mask of VLAN 2] → Next.
44. Router: [gateway IP of VLAN 2] → Add.
45. DNS Server: [server IP of VLAN 2] → Add.
46. Activate → Finish.

Step B.4: Test DHCP from a Client

47. On a client laptop connected to the switch, set the NIC to obtain IP automatically.
48. Run: `ipconfig /release` then `ipconfig /renew`
49. Confirm the client received an IP in the expected DHCP pool range.
50. Ping the server IP from the client to verify reachability.

Task C — Install and Configure DNS Server

Step C.1: Install the DNS Server Role

51. Server Manager → Manage → Add Roles and Features.
52. Select DNS Server → Add Features → Install.
53. Wait for completion, then close the wizard.

Step C.2: Create a Forward Lookup Zone

54. Server Manager → Tools → DNS.
55. Expand your server → right-click Forward Lookup Zones → New Zone.
56. Zone Type: Primary zone → Next.
57. Zone name: group[N].lab (replace N with your group number, e.g., group1.lab).
58. Accept defaults for zone file → Next → Finish.

Step C.3: Add a Host (A) Record for the Server

59. Expand Forward Lookup Zones → click your new zone (group[N].lab).
60. Right-click in the right pane → New Host (A or AAAA).
61. Name: server
62. IP address: [server IP of your VLAN 1]
63. Check Create associated pointer (PTR) record.
64. Click Add Host → OK.
65. Repeat and add a second A record for VLAN 2 server IP if needed.

Step C.4: Test DNS Resolution

66. From the server itself, open CMD and run: nslookup server.group[N].lab
67. Expected result: returns the server IP you configured.
68. From a client (after receiving DHCP): run the same nslookup.
69. If not resolving, ensure DHCP option 006 points to the server IP.

NOTE

If nslookup shows 'Non-existent domain', right-click your zone → Reload, then retry.

Task D — Install IIS and Publish a Web Page

Step D.1: Install the IIS Role

70. Server Manager → Manage → Add Roles and Features.
71. Select Web Server (IIS) → Add Features → Next → Next → Install.
72. Wait for completion, then close.

Step D.2: Verify IIS Default Site

73. Open Internet Explorer or Edge on the server.
74. Navigate to http://localhost
75. You should see the IIS Welcome page.
76. From a client machine, navigate to http://[server IP] — same page should appear.

Step D.3: Create a Custom Default Page

77. Open File Explorer → navigate to C:\inetpub\wwwroot\
78. Right-click → New → Text Document → rename it to index.html (confirm the extension change).
79. Open index.html with Notepad and type the following content:
80. <html><body><h1>Group [N] Lab Server</h1><p>VLAN [ID1] and VLAN [ID2] - All services running.</p></body></html>
81. Save and close the file.
82. Refresh the browser at http://localhost — you should see your custom page.
83. From the client, navigate to http://[server IP] to confirm.

NOTE

Replace [N], [ID1], [ID2] with your actual group number and VLAN IDs.

Step D.4: Test from Both VLANs

- 84. From a VLAN 1 client: open browser → http://[server IP VLAN 1] → page loads.
- 85. From a VLAN 2 client: open browser → http://[server IP VLAN 2] → page loads.
- 86. Also test using DNS name: http://server.group[N].lab

Task E — Install Print Services and Add a Virtual Printer**Step E.1: Install the Print and Document Services Role**

- 87. Server Manager → Manage → Add Roles and Features.
- 88. Select Print and Document Services → Add Features.
- 89. Ensure Print Server sub-role is checked.
- 90. Click Next → Next → Install → Close when done.

Step E.2: Add a Printer (Virtual / Generic)

- 91. Server Manager → Tools → Print Management.
- 92. Expand Print Servers → your server name → right-click Printers → Add Printer.
- 93. Select Add a new printer using an existing port → use LPT1: (virtual) → Next.
- 94. Choose Install a new driver → Next.
- 95. Manufacturer: Generic | Printer: Generic / Text Only → Next.
- 96. Printer name: Group[N]-VirtualPrinter
- 97. Check Share this printer and enter share name: Group[N]-Print
- 98. Click Next → Finish.

Step E.3: Connect a Client to the Shared Printer

- 99. On the client machine, open Run (Win+R) → type \\[server IP]\Group[N]-Print → Enter.
- 100. Windows will install the printer driver automatically.
- 101. Open any document → File → Print → select Group[N]-VirtualPrinter.
- 102. Print a test page to confirm.

NOTE

If the client cannot reach the share, verify Windows Firewall on the server allows File and Printer Sharing.

Section 5 — Final Verification Checklist

Before calling the instructor, verify all items below. Your peer group should cross-check your results.

Task	Verified by student	Verified by peer	Instructor notes
Static IP assigned & ping OK			
DHCP scope active (both VLANs)			
Client receives IP from DHCP			

Task	Verified by student	Verified by peer	Instructor notes
DNS zone created & resolving			
IIS default page loads in browser			
Virtual printer installed & shared			
Test print page printed			

GRADING NOTE

Each completed and verified task earns equal marks.
 Both VLANs must be functional — partial credit applies per VLAN.
 Clean IP plan and proper DHCP pool boundaries will be checked.

Section 6 — Quick Troubleshooting Reference

Problem	Solution
Client does not receive an IP	Check DHCP scope is active. Verify DHCP server NIC is on the same VLAN as the client. Ensure no exclusion covers the pool. Restart DHCP service.
Ping between client and server fails	Check physical connection to switch. Confirm VLAN tagging on switch port. Verify server static IP is in the correct subnet. Disable Windows Firewall temporarily for testing.
nslookup fails on client	Confirm DHCP option 006 (DNS) is set to the server IP. Restart DNS Client service on the client. Verify DNS forward zone name matches what you are querying.
IIS page not loading from client	Check IIS is running (services.msc). Ensure Windows Firewall allows port 80. Try http://[IP] not https://[IP]. Verify index.html is saved in C:\inetpub\wwwroot\.
Cannot connect to shared printer	Enable File and Printer Sharing in Windows Firewall. Confirm printer is shared. Try connecting via UNC path: \\[IP][sharename]. Ensure client has network access to server.

Student Notes

Use this space to record commands, IP addresses, observations, and any issues encountered.